- **Partitioning clustering**, aka *flat clustering*, outputs a division of the input dataset into partitions, and so no cluster is the subset of another one, nor does it intersect any other cluster.
- **Hierarchical clustering**, instead, produces a hierarchy of clusters that can be then interpreted and "sliced" at any given point, depending on parameters set for the algorithm.

Table 12.1 Summary of the characteristics of clustering algorithms presented in this chapter.

Categories	K-means	DBSCAN	OPTICS
Membership	Hard	Hard	Soft
Structure	Flat	Flat	Hierarchical
Strategy	Centroid-based	Density-based	Density-based
Determinism	Randomized	Deterministic	Deterministic
Outliers detection	No	Yes	Yes

Obviously these 2 criteria are orthogonal, so for instance you can have hard partitioning clustering algorithms (like *k-means*) or soft hierarchical ones (like *OPTICS*).

Other criteria often used to classify these algorithms are: **centroid-based** versus **density-based**, and **randomized** versus **deterministic**.

In the next section, we'll first describe an algorithm for partitioning clustering, k-means, the progenitor of all clustering algorithms. Then we move to a different type of flat clustering, DBSCAN, and finally we'll pick up the discussion on hierarchical clustering, and introduce OPTICS,